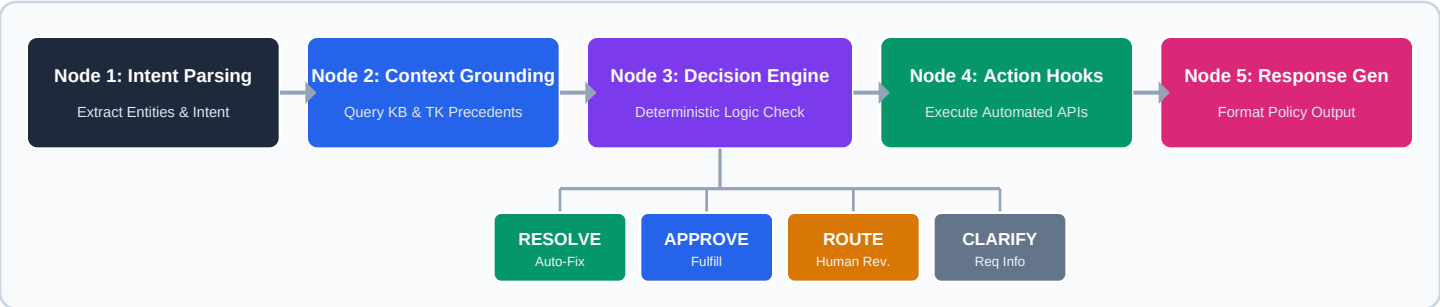


Veridian Corp — IT Support Agent Architecture

Deterministic-First Hybrid Control Pipeline | Antigravity Framework Blueprint | Period: 21–25 Sep 2026

1. High-Level Control System Architecture

The Veridian IT Support Agent operates on a **Deterministic Hybrid Control Pipeline**. To ensure 100% policy compliance and zero hallucination, LLMs are restricted from making ungrounded decisions. The LLM parses natural language input into structured entities, while a deterministic Python State Engine evaluates company policies (KB-01 to KB-10, Asset Policy) and ticket precedents (TK-1042 to TK-1051) to execute decisions.



Pipeline Stage	Function & Operational Responsibility
Node 1: Intent & Entity Parsing	Extracts intent category, device age, error logs, user credentials, and extra metadata.
Node 2: Context & Policy Grounding	Matches extracted parameters against Knowledge Base rules (KB-01 to KB-10) and historical ticket precedents.
Node 3: Policy Decision & Audit Trail	Evaluates rules deterministically to assign System Outcome State: RESOLVE, APPROVE, ROUTE, or CLARIFY.
Node 4: Action Execution Hooks	Triggers automated backend functions (Account Unlock, VPN Renewal, Ticket Escalation).
Node 5: Response Generation	Formats clear, empathetic, policy-grounded response citing specific rules and precedent tickets.

2. Core Unified JSON State Schema

```
{ "request_id": "REQ-10", "employee": { "name": "Kavya Pillai", "role": "Senior Accountant", "is_contractor": false }, "intent_category": "ACCESS_ADMIN", "extracted_entities": { "target_system": "finance_reporting_server", "access_level": "admin" }, "matched_kb_rules": [ "KB-04" ], "matched_precedents": [ { "ticket_id": "TK-1050", "status": "PENDING_APPROVAL" } ], "decision": { "status": "ROUTE", "action_type": "ROUTE_FOR_APPROVAL_AND_JUSTIFICATION", "routing_target": "FINANCE_SECURITY_MANAGER" }, "audit_trail": { "policy_reasoning": "Admin access request requires business justification per KB-04 (Precedent TK-1050)." } }
```

3. Decision Matrix & Action Taxonomy

Intent Category	Matched Policy / Rule	System State	Target Action Hook
Password Lockout	KB-01 (≥5 failed attempts)	RESOLVE	unlock_user_account()
VPN (Full-Time / Contractor)	KB-02 (Auto-grant vs Mgr Approval)	RESOLVE / ROUTE	renew_vpn() / route_manager()
Laptop Replacement	KB-03 / Asset Policy (≥3 yrs / Failure)	APPROVE / ROUTE	dispatch_laptop() / route_repair()
Non-Catalog Software	KB-04 (3-5 day Security review)	ROUTE	escalate_security_review()
Printer Malfunction	KB-05 (Restart spooler + Asset Tag)	ROUTE	dispatch_field_support()
Mailbox Quota	KB-06 (25GB default, max 50GB)	ROUTE	request_quota_approval()
Guest Wi-Fi	KB-07 (24-hr self-service kiosk)	RESOLVE	provide_kiosk_instructions()
Expense Portal	KB-08 (Finance DB account lookup)	ROUTE	redirect_to_finance()
Phishing Threat	KB-09 (Quarantine + No-Forward)	ROUTE	escalate_phishing_alert()
WFH Equipment	KB-10 (>3 days remote + Sign-off)	ROUTE	route_finance_manager()
Ambiguous Input	Baseline Security Protocol	CLARIFY	request_clarification()